

D863

Waves, Acoustics, and Sound

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A science teacher, whose classroom speaker was donated by a music store, plays a 440 Hz tone and then an 880 Hz tone in the same room at 20 °C. The speaker is 1.2 m from the front row. How do the speed and wavelength of the 880 Hz tone compare with those of the 440 Hz tone?

- A. Same speed, twice the wavelength.
- B. Twice the speed, same wavelength.
- C. Same speed, half the wavelength.
- D. Speed and wavelength both unchanged.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A teacher demonstrates with a long spring toy bought at a toy store. Pushing and pulling one end along the spring's length sends compressions down it. The spring is 2 m long when stretched. Which statement is correct about this wave and about sound?

- A. It is longitudinal, like sound, so neither can be polarized.
- B. All mechanical waves are longitudinal.
- C. It is transverse, like sound.
- D. It is longitudinal, unlike sound, which is transverse.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

Two speakers in a school auditorium with 400 seats play the same 340 Hz tone in phase. Sound travels at 340 m/s. A listener sits 5.5 m from one speaker and 6.0 m from the other. The auditorium was renovated last year. What does the listener hear?

- A. A quiet spot, because the path difference is half a wavelength (destructive).
- B. A quiet spot, because 6.0 m is far from a speaker.
- C. Beats at 0.5 per second.
- D. A loud spot, because the waves arrive in phase.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Two in-phase speakers emit sound with a wavelength of 0.80 m. What is the smallest nonzero path difference that gives constructive interference?

- A. 0.40 m
- B. 0.80 m
- C. 1.60 m
- D. 0.20 m

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A band director tuning with a 440 Hz tuning fork bought last fall hears 4 beats per second against a clarinet's A. The clarinet is made of grenadilla wood. When the player raises the pitch slightly, the beats speed up. What was the clarinet's original frequency?

- A. 436 Hz
- B. Either 436 or 444 Hz; there is no way to tell.
- C. 440 Hz, since the beats come from the room.
- D. 444 Hz

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Two tuning forks, 256 Hz and 262 Hz, are sounded together. What beat frequency is heard?

- A. 259 Hz
- B. 3 Hz
- C. 6 Hz
- D. 518 Hz

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A music teacher builds a pan-flute tube, closed at the bottom and 0.25 m long, from bamboo harvested at a local farm. Sound travels at 340 m/s. The tube is painted in bright colors. What is its fundamental frequency?

- A. 340 Hz
- B. 680 Hz
- C. 85 Hz
- D. 1,360 Hz

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A pipe closed at one end has a fundamental frequency of 200 Hz. What is its next resonant frequency?

- A.** 100 Hz
- B.** 600 Hz
- C.** None; a closed pipe has only one resonance.
- D.** 400 Hz

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

At an outdoor concert on a fairground rented for the weekend, a sound meter 10 m from a speaker reads an intensity of 0.080 W/m^2 . The speaker cabinet weighs 45 kg. Assuming open field and no reflections, what is the intensity 40 m away?

- A.** 0.020 W/m^2
- B.** 0.005 W/m^2
- C.** 0.080 W/m^2 , since outdoor sound does not weaken.
- D.** 1.28 W/m^2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A listener wants the sound intensity from a speaker to fall to one-ninth of its current value. By what factor must she increase her distance?

- A.** 9 times
- B.** $1/3$ as far
- C.** 81 times
- D.** 3 times

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A lesson uses an online simulation (free for teachers) showing a 500 Hz sound wave passing from air into water, where it travels about 4.3 times faster. The simulation has a slider for color themes. What happens to the wave's frequency and wavelength in the water?

- A.** Both stay the same.
- B.** The frequency rises 4.3 times, and the wavelength stays the same.
- C.** The frequency stays 500 Hz, and the wavelength becomes about 4.3 times longer.
- D.** The frequency stays the same, and the wavelength gets shorter.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

At a music festival with 12 performing groups, a violin and a flute play the same note at the same loudness, yet listeners easily tell them apart. The festival runs from 9 a.m. to 3 p.m. What property differs?

- A.** Their amplitude.
- B.** Their fundamental frequency.
- C.** Nothing physical; the difference is only in the listener's mind.
- D.** Timbre — the mix of harmonics each instrument produces.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A science teacher, ~~whose classroom speaker was donated by a music store~~, plays a 440 Hz tone and then an 880 Hz tone in the same room at 20 °C. ~~The speaker is 1.2 m from the front row.~~ How do the speed and wavelength of the 880 Hz tone compare with those of the 440 Hz tone?

Target word: speed and wavelength when frequency doubles **Decides:** Same medium; v fixed.

A	Inverted operation	Higher frequency means shorter wavelength.
B	Confusion set	Frequency does not set the speed of sound.
C	Correct	v is set by the air; $\lambda = v/f$ halves when f doubles.
D	Dropped element	Drops the link between frequency and wavelength.

2. Answer A Concept 2

A teacher demonstrates with a long spring toy ~~bought at a toy store~~. Pushing and pulling one end along the spring's length sends compressions down it. ~~The spring is 2 m long when stretched.~~ Which statement is correct about this wave and about sound?

Target word: wave type **Decides:** Vibration along the direction of travel.

A	Correct	Particles move back and forth along the direction the wave travels.
B	Absolute language	Waves on a string and water-surface waves are transverse.
C	Confusion set	Sound in air is longitudinal.
D	Inverted operation	Sound is longitudinal too.

3. Answer A Concept 3

Two speakers in a school auditorium ~~with 400 seats~~ play the same 340 Hz tone in phase. Sound travels at 340 m/s. A listener sits 5.5 m from one speaker and 6.0 m from the other. ~~The auditorium was renovated last year.~~ What does the listener hear?

Target word: interference at the seat **Decides:** $\lambda = 1.0$ m; path difference = 0.5 m.

A	Correct	Half a wavelength puts the waves out of phase.
B	Dropped element	Right result, but it drops the path-difference reasoning that decides it.
C	Confusion set	Beats need two different frequencies; both speakers play 340 Hz.
D	Inverted operation	A half-wavelength difference puts them out of phase.

4. Answer B Concept 3 · transfer

Two in-phase speakers emit sound with a wavelength of 0.80 m. What is the smallest nonzero path difference that gives constructive interference?

Target word: path difference for constructive interference **Decides:** One whole wavelength.

A	Inverted operation	Half a wavelength gives destructive interference.
B	Correct	The first nonzero whole-wavelength difference is 1λ .
C	Wrong stem number	That is 2λ — constructive, but not the smallest.
D	Confusion set	A quarter wavelength is neither fully constructive nor destructive.

5. Answer D Concept 4

A band director tuning with a 440 Hz tuning fork ~~bought last fall~~ hears 4 beats per second against a clarinet's A. ~~The clarinet is made of grenadilla wood.~~ When the player raises the pitch slightly, the beats speed up. What was the clarinet's original frequency?

Target word: original frequency **Decides:** 4 beats and faster beats when raised.

A	Inverted operation	Raising 436 Hz toward 440 would slow the beats.
B	Dropped element	Drops the clue that raising pitch sped up the beats.
C	Confusion set	Beats come from two different frequencies, not room echoes.
D	Correct	It is 436 or 444 Hz; raising the pitch increases the gap only if it was already above 440.

6. Answer C Concept 4 • transfer

Two tuning forks, 256 Hz and 262 Hz, are sounded together. What beat frequency is heard?

Target word: beat frequency **Decides:** $|f_1 - f_2|$.

A	Confusion set	That is the average — the pitch heard, not the beat rate.
B	Wrong stem number	Halves the difference.
C	Correct	$262 - 256 = 6$ beats per second.
D	Inverted operation	Adds instead of subtracting.

7. Answer A Concept 5

A music teacher builds a pan-flute tube, closed at the bottom and 0.25 m long, ~~from bamboo harvested at a local farm.~~ Sound travels at 340 m/s. ~~The tube is painted in bright colors.~~ What is its fundamental frequency?

Target word: fundamental of a closed tube **Decides:** $\lambda = 4L = 1.0$ m.

A	Correct	$f = 340 \div 1.0 = 340$ Hz.
B	Confusion set	Uses $\lambda = 2L$, the rule for a pipe open at both ends.
C	Inverted operation	Multiplies 340×0.25 instead of dividing by $4L$.
D	Dropped element	Uses $\lambda = L$, dropping the factor of 4.

8. Answer B Concept 5 • transfer

A pipe closed at one end has a fundamental frequency of 200 Hz. What is its next resonant frequency?

Target word: next harmonic of a closed pipe **Decides:** Only odd harmonics.

A	Inverted operation	Resonances lie above the fundamental, not below.
B	Correct	Closed pipes resonate at 1, 3, 5 ... times the fundamental.
C	Absolute language	Closed pipes have a whole series of odd harmonics.
D	Confusion set	The second harmonic exists only in pipes open at both ends.

9. Answer B Concept 6

At an outdoor concert on a fairground rented for the weekend, a sound meter 10 m from a speaker reads an intensity of 0.080 W/m^2 . ~~The speaker cabinet weighs 45 kg.~~ Assuming open field and no reflections, what is the intensity 40 m away?

Target word: intensity at a new distance **Decides:** 4 times the distance $\rightarrow 1/16$.

A	Dropped element	Divides by 4, dropping the square.
B	Correct	$0.080 \div 4^2 = 0.080 \div 16 = 0.005 \text{ W/m}^2$.
C	Absolute language	Sound spreads over a larger area and always weakens with distance.
D	Inverted operation	Multiplies by 16; intensity falls with distance.

10. Answer D Concept 6 · transfer

A listener wants the sound intensity from a speaker to fall to one-ninth of its current value. By what factor must she increase her distance?

Target word: distance factor from intensity **Decides:** $I \propto 1/r^2$.

A	Dropped element	Drops the square root.
B	Inverted operation	Moving closer raises intensity.
C	Confusion set	Squares the factor instead of taking its root.
D	Correct	$(1/3)^2 = 1/9$.

11. Answer C Concept 7

A lesson uses an online simulation (~~free for teachers~~) showing a 500 Hz sound wave passing from air into water, where it travels about 4.3 times faster. ~~The simulation has a slider for color themes.~~ What happens to the wave's frequency and wavelength in the water?

Target word: frequency and wavelength across a boundary **Decides:** Frequency is fixed by the source.

A	Dropped element	Drops the change in speed.
B	Confusion set	Frequency is set by the source; it does not change at a boundary.
C	Correct	$\lambda = v/f$; speed rises, frequency does not.
D	Inverted operation	Faster speed with fixed frequency means longer wavelength.

12. Answer D Concept 8

At a music festival with 12 performing groups, a violin and a flute play the same note at the same loudness, yet listeners easily tell them apart. ~~The festival runs from 9 a.m. to 3 p.m.~~ What property differs?

Target word: what makes them sound different **Decides:** Same pitch and loudness.

A	Adjacent category	Same loudness means similar amplitude.
B	Confusion set	Same note means same fundamental — same pitch.
C	Absolute language	The waveforms really differ and can be measured.
D	Correct	Different harmonic content gives each a different waveform shape.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	1	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	11	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	11	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	The medium sets wave speed	Wave speed depends on the medium, not the frequency; in one medium, raising frequency shortens wavelength ($v = f\lambda$).	Thinking higher-pitched sound travels faster.	Two frequencies in the same medium.
2	Transverse vs. longitudinal waves	In longitudinal waves (sound) particles vibrate along the direction of travel; in transverse waves they vibrate across it. Only transverse waves can be polarized.	Calling sound a transverse wave.	A demo that pushes and pulls along a spring.
3	Interference from path difference	For in-phase sources, a path difference of a whole number of wavelengths gives constructive interference; an odd half-wavelength gives destructive interference.	Judging loudness by distance to the nearest speaker.	Two distances to two speakers and a wavelength (or f and v).
4	Beats	Two close frequencies produce beats at $ f_1 - f_2 $ per second; beats speeding up as one pitch rises means that pitch was already above the other.	Adding or averaging the frequencies to get the beat rate.	A tuning fork, a beat rate, and a pitch adjustment.
5	Resonance in pipes	A pipe closed at one end has a fundamental wavelength of $4L$ and only odd harmonics; a pipe open at both ends has $2L$ and all harmonics.	Using the open-pipe rule for a closed tube.	A tube length and whether its ends are open or closed.
6	Inverse-square intensity	From a point source in open space, intensity falls as $1/r^2$: doubling distance gives one-quarter the intensity.	Using $1/r$ (halving) instead of $1/r^2$.	Intensity at one distance and a question at another.
7	Frequency is fixed at a boundary	When a wave enters a new medium, its frequency stays the same; its speed and wavelength change together.	Thinking frequency changes when speed changes.	A wave crossing from one medium to another.
8	Pitch, loudness, and timbre	Pitch follows frequency, loudness follows amplitude (intensity), and timbre (tone quality) follows the mix of harmonics.	Explaining different-sounding instruments by pitch or loudness.	Same note, same loudness, different sound.

Course scope (public description): Waves, Acoustics, and Sound provides a comprehensive exploration of fundamental wave properties and their interactions with various media. Designed for learners preparing for roles in science education, it delves into the core concepts of wave behavior, including wave characteristics, factors affecting wave propagation, and the effects of interference. Through detailed lessons on wave properties, interactions, and changes in wave behavior, learners will gain the skills needed to analyze and predict wave phenomena. The course emphasizes practical applications, employing interactive simulations and real-world examples to enhance understanding and prepare learners for effective teaching and application of wave concepts in educational settings.